Science and Engin...
SCHOOL OF EN...
Concept Mapping...

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1.
(a) · To acquire images from the real world.
· To process and analyze visual data.
· To recognize and classify objects.

(b) 1. Healthcare (medical image diagnosis)
2. Autonomous vehicles (lane and obstacle detection)

2.
(a) Feature extraction is the process of identifying important characteristics from an image to simplify analysis and object recognization.
(b) Object recognization enables the system to identify objects accurately, improving automation, decision-making.

3.
(a) Low-level image processing performs basic operations such as image enhancement, noise removal, filtering and contrast without interupting image content.

(b)

Mid-level image processing	High-level image processing
1. performs segmentation and feature extraction.	1. performs object recognition and scene understanding.
2. identifies regions or objects.	2. Make intelligent decision based on image.

4.
(a) pixel is a smallest unit of image that stores intensity or colour info.
(b) High resolution contains more pixels, providing more detail and better image quality.

5.
(a) Image sensor captures light captures light from a scene and conv... it into electrical signals.

(b) 1. Digital camera.
2. scanner.

6.
(a) Sampling is the process of converting a continuous image into a grid of discrete pixels.

(b) Quantization assigns discrete intensity values to sampled pixels.

7.
(a) prespective projection is the process of projecting a 3D scene onto a 2D image plane, while preserving depth perception.

(b) illumination provides the light reflected from objects, making them visible for image capture and influencing image brightness & Quality.

8.
(a) Image processing is the technique of enhancing, restoring images.

(b)

image processing	Computer vision
1. Enhances images	1. Enterprets & Understands images.
2. Output is an improved image.	2. Output is information or decisions.

9.
(a) Computer vision analyzes medical images such as X-rays, CT scans, MRI.
(b) Computer vision detects lanes, signs, pedestrains and obstacles.

10.
(a) pixels store the colour & itensity infoo that forms complete image.
(b) sampling determines the number of pixels, while higher image resolution provides more detail, leading to more accurate extracting.